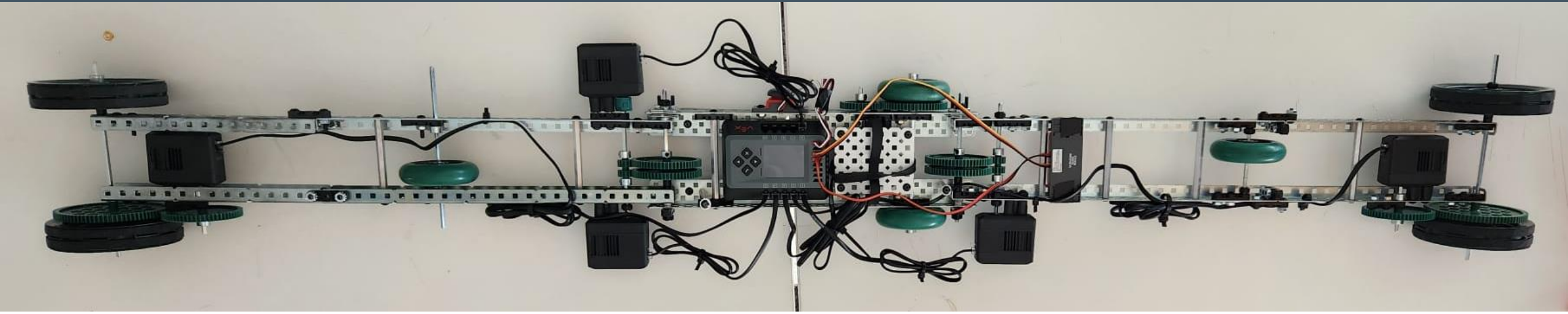




Team Woman in STEM



Propulsive Mechanical Spanner (PMS)

2026 UBC Senior Engineering Design Competition

HATCH

Management Team Structure

Software Lead and Woman in STEM:

Emma

Mechanical Lead:

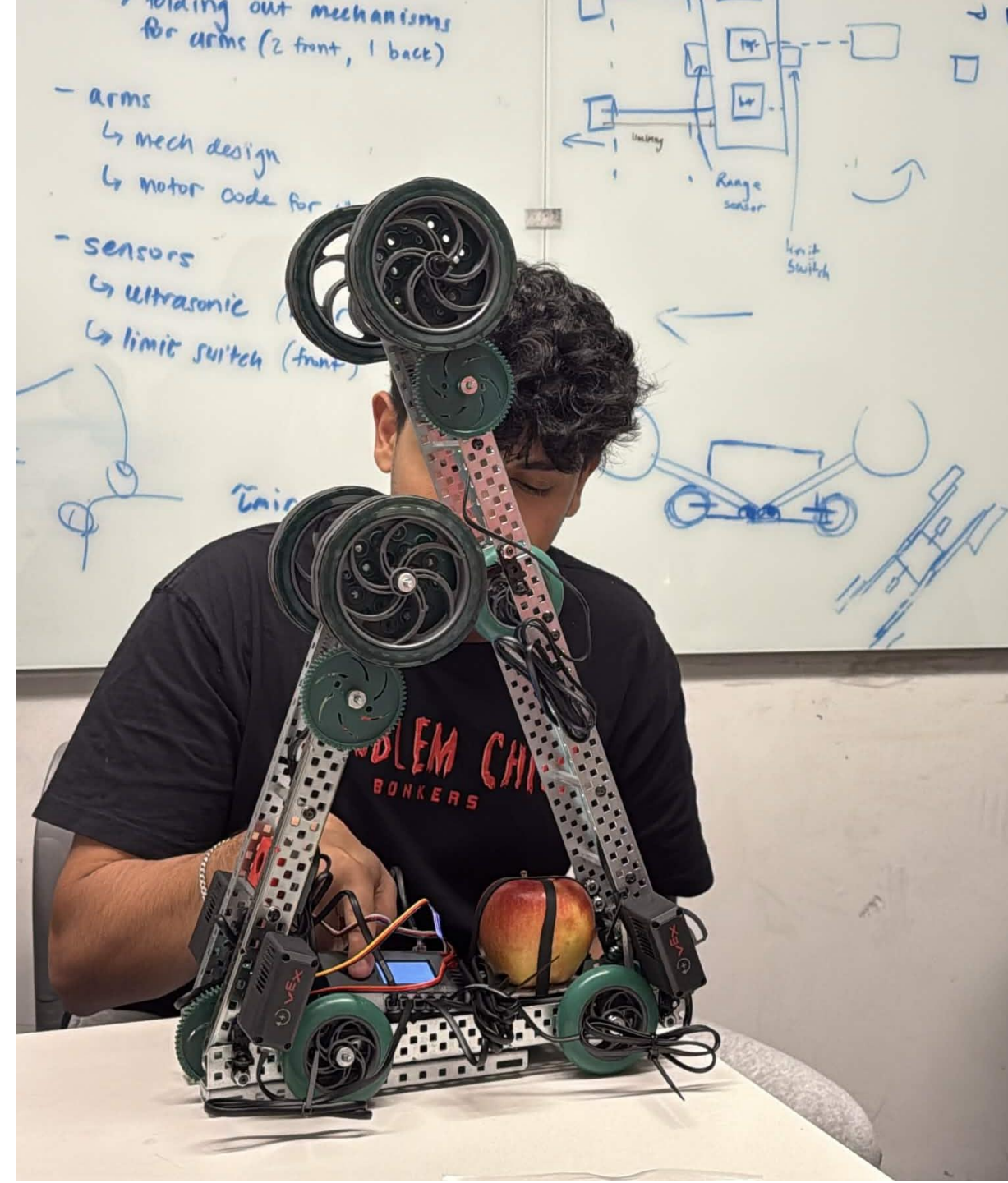
Michael, Dom

Instrumentation and Integration Lead:

Arman

Mechanical / Documentation:

Nick



Management

The Team

- Distributed, collaborative team
 - Group planning, ideation and problem-solving
 - Parallel task completion
 - Sharing prior knowledge



Design Process

Concept Generation

12:00-12:30

Competition Rule Overview

12:30-12:45

Individual Concept Generation / Brainstorming

12:45

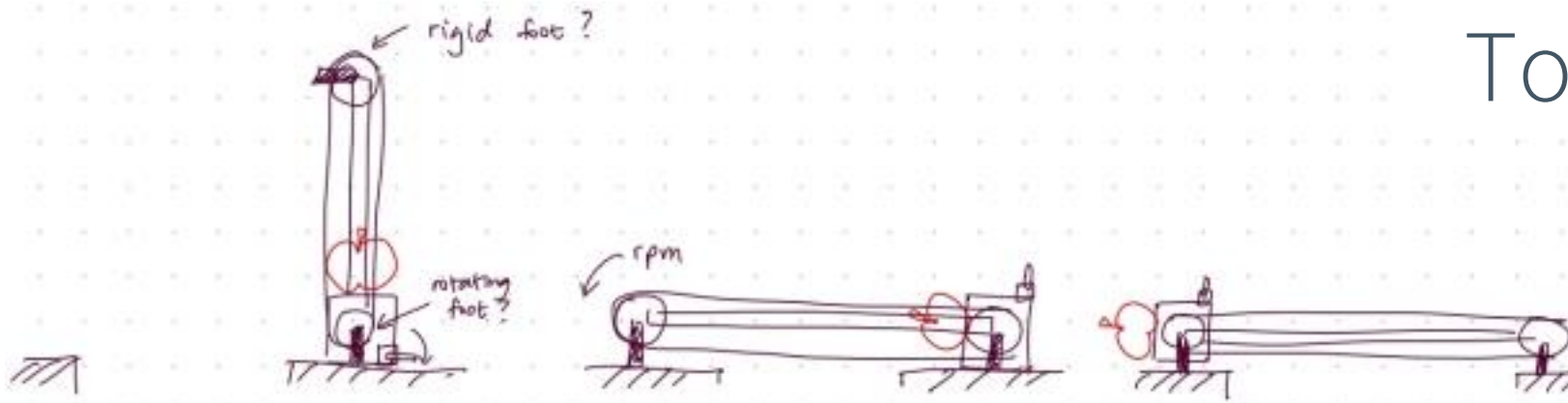
Concept Evaluation / Equipment Testing

1:30-2:00

Lunch

Individual Concept Generation

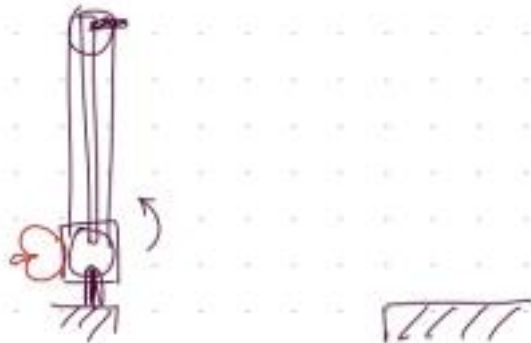
Topple Bot



1. servo rotate to topple
↳ or controlled lowering of
track since it has to
be pulled back up

2. chain drive on

3. payload across



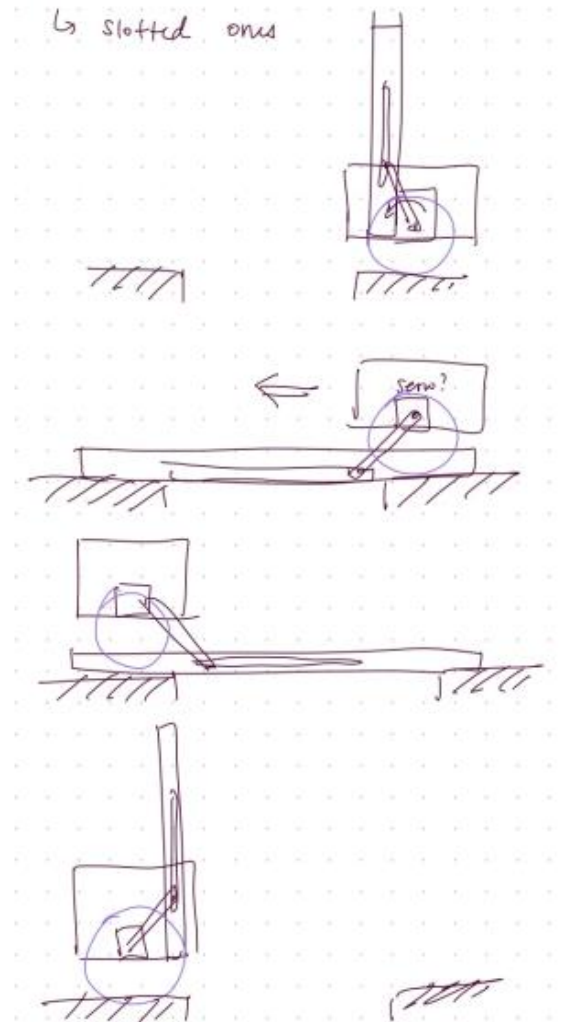
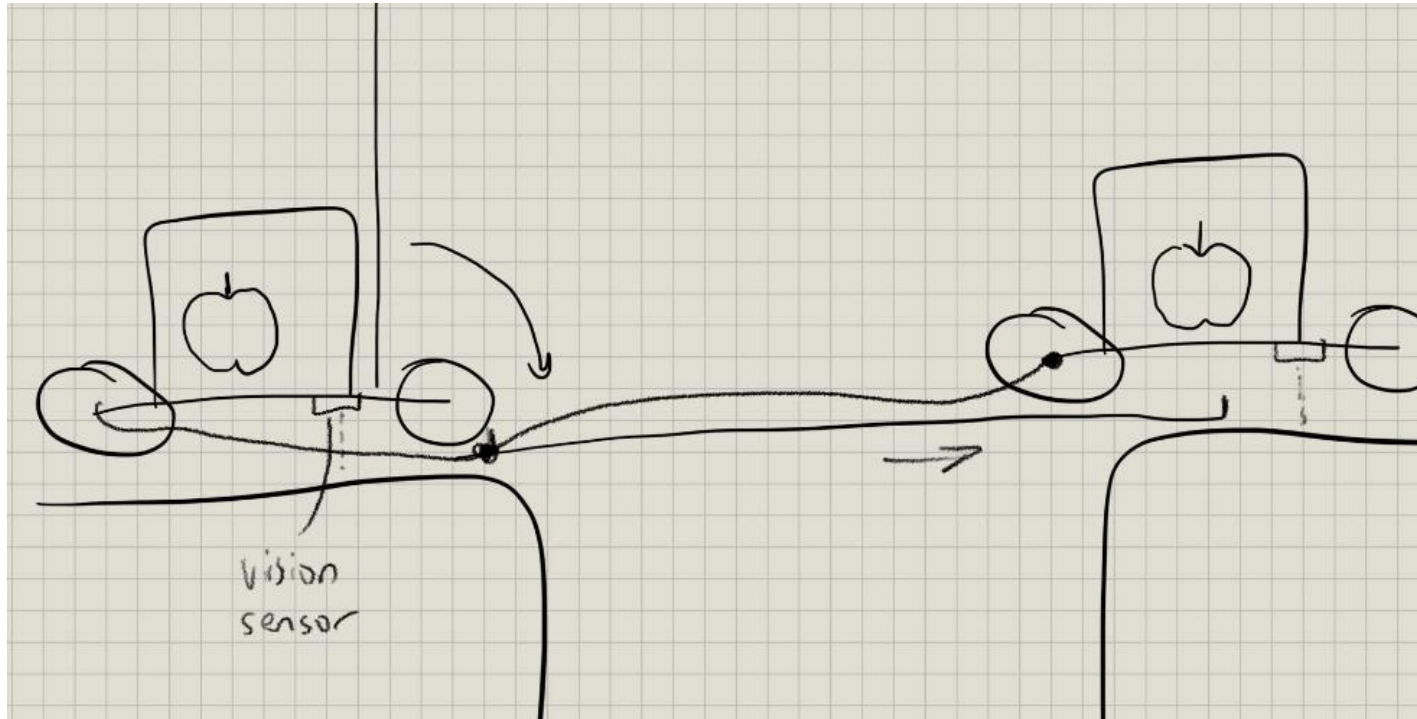
4. track pulled up
Done!

- gap span limited
by track piece

- chain drive gears
are separated from
the brain/payload
carriage

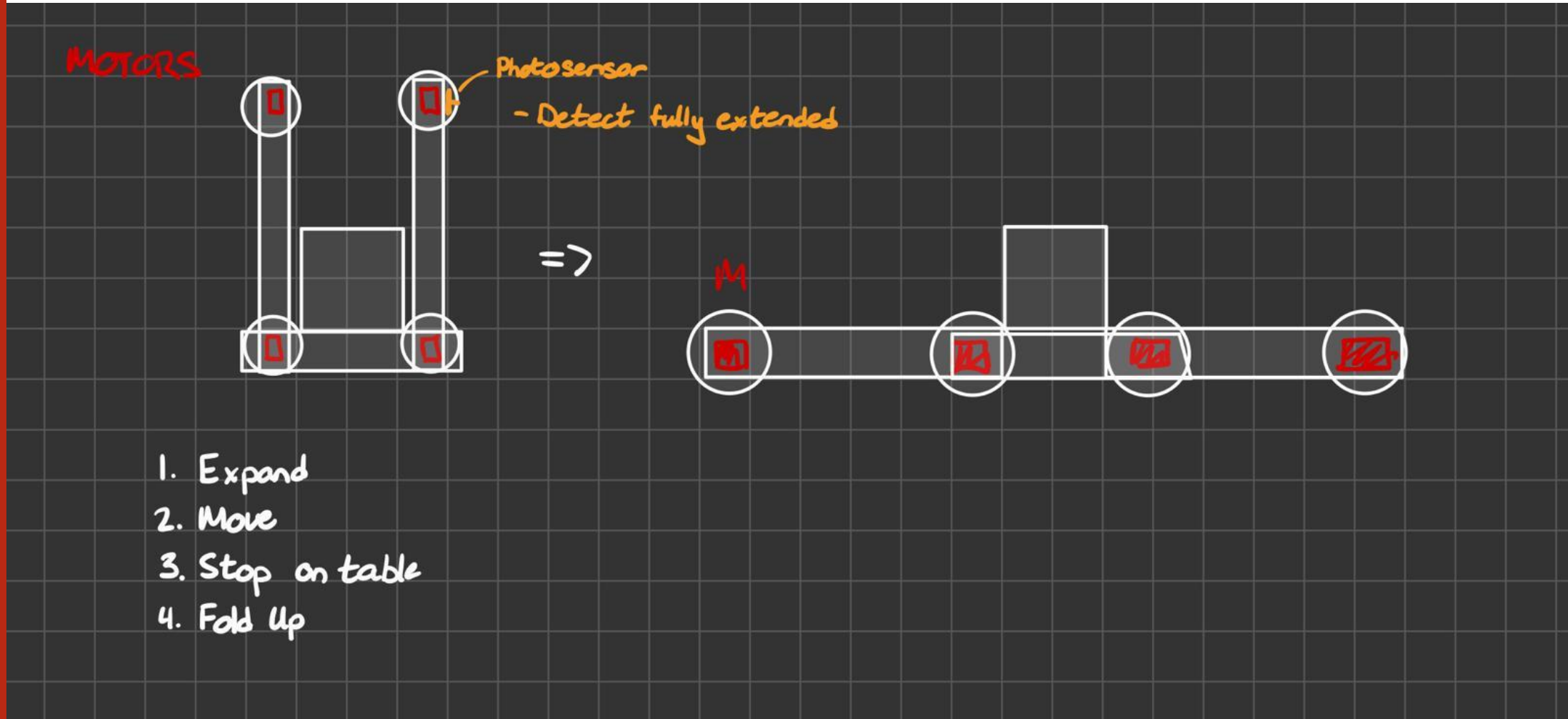
Individual Concept Generation

Track Laying Bot



Individual Concept Generation

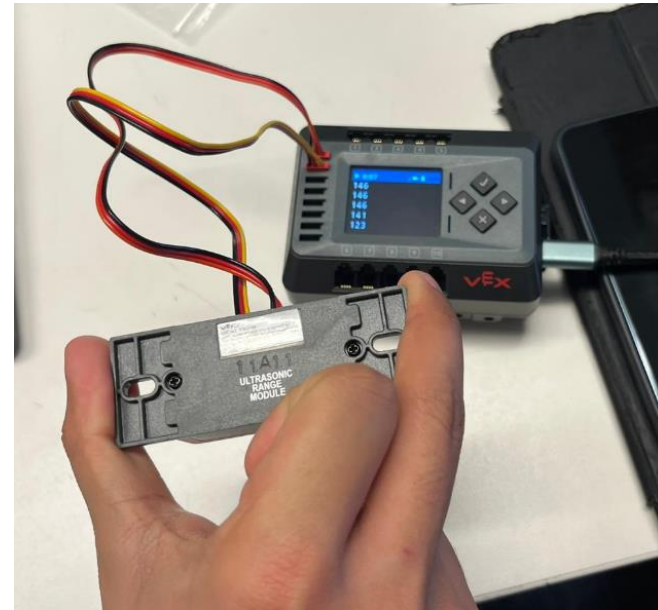
Wiper Bridge Bot



Design Process

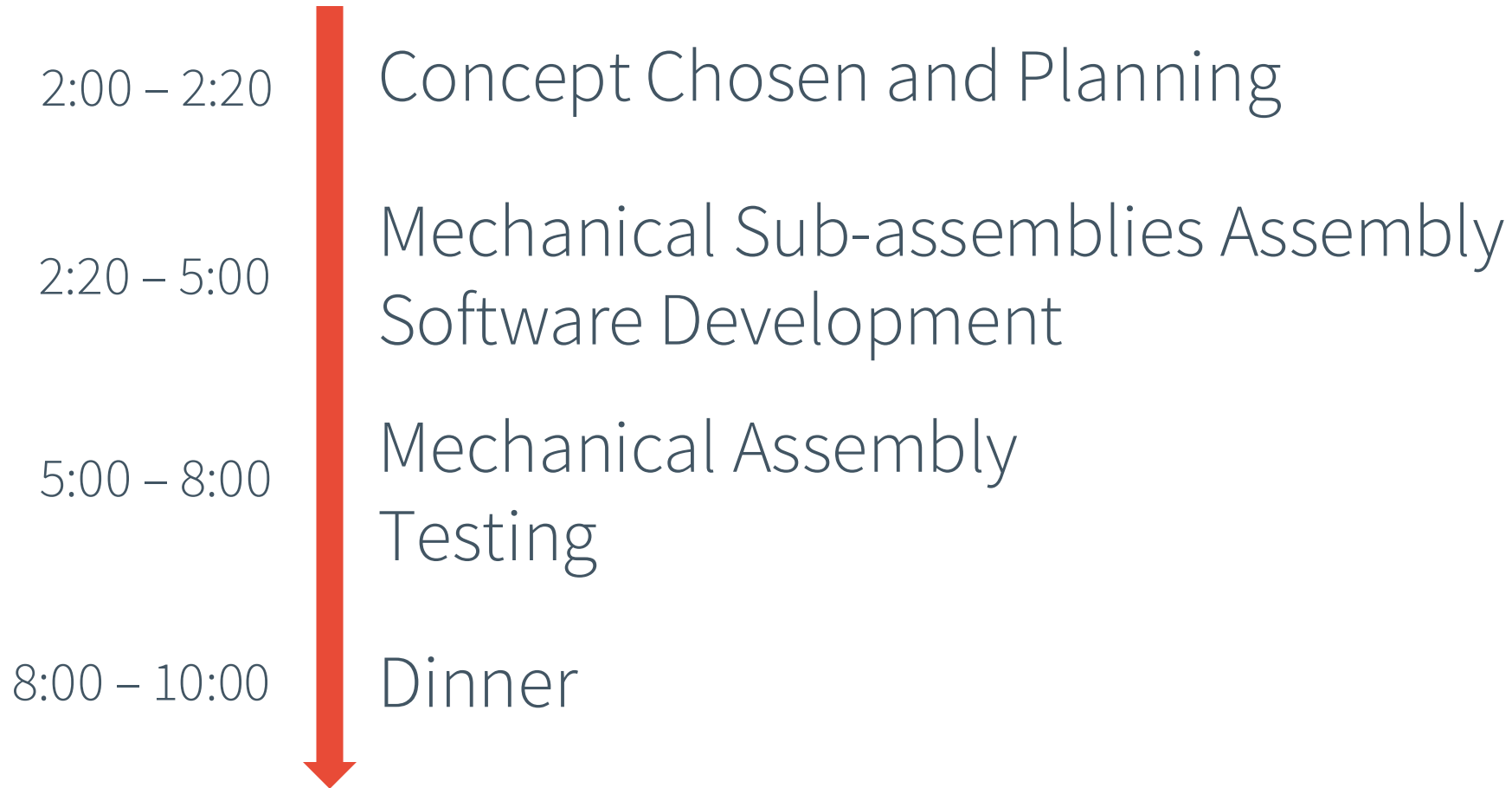
Concept Selection

- Eliminate ideas based on feasibility:
 - Eliminated Topple Bot because no chain drive
- Checked motor torque and ultrasonic sensor function



Design Process

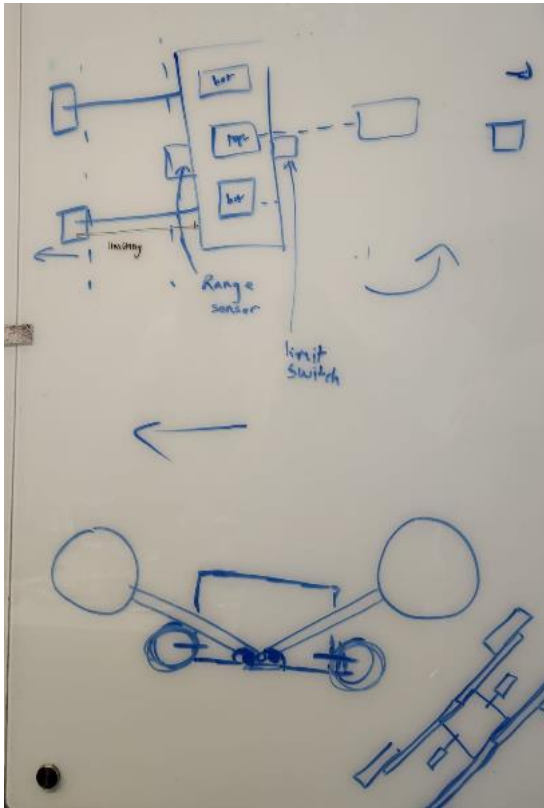
Concept Generation



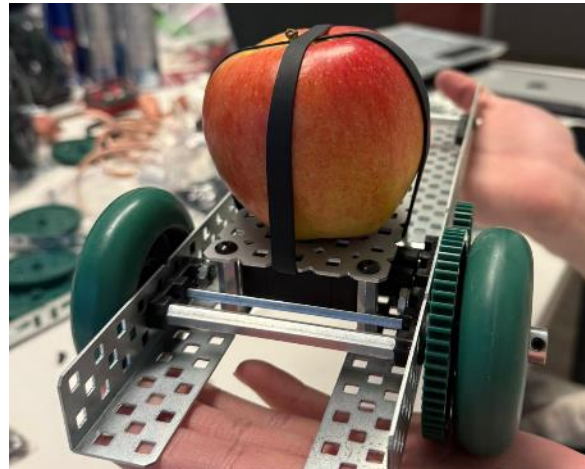
Design Process

Planning + Sub-assembly Development

Team unanimously agreed to proceed with Wiper Bridge concept



Detailed sketch



Mechanical Breakdown

Software

- pushbutton press
- rear arm go down
- rear motor drive until limit switch released
- rear motor stop
- front arms go down
- all motors drive forward until ultrasonic sensor detects table + buffer
- all motors brake
- all arms return to upright

when BumperH pressed

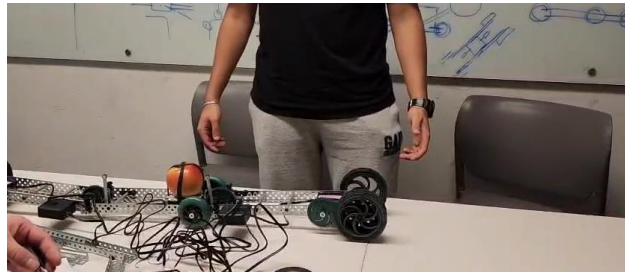
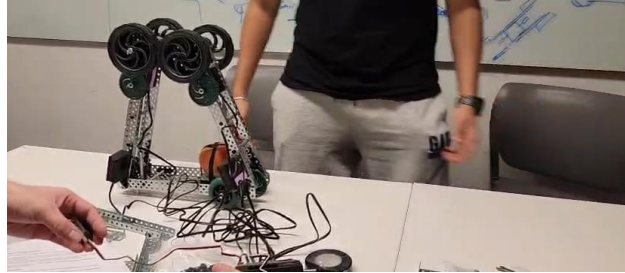
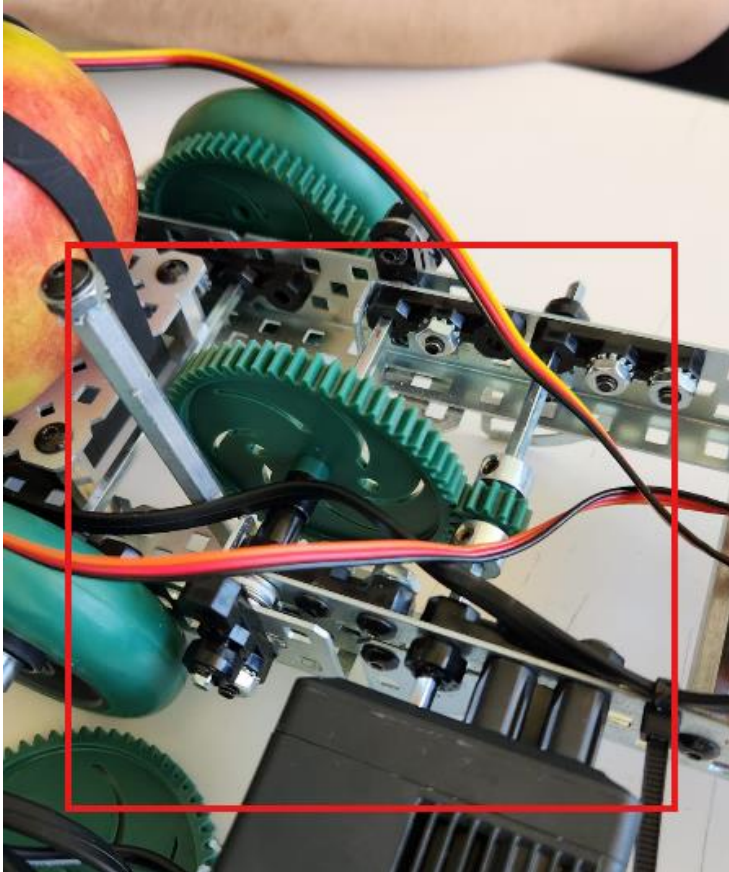
- set RearArmPos max torque to 100 %
- set RearArmPos2 max torque to 100 %
- set FrontArmPos max torque to 100 %
- set RearArmPos velocity to 25 %

Software Breakdown

Design Process

Mechanical Assembly + Testing

Design of arm rotator, testing to get gear ratio

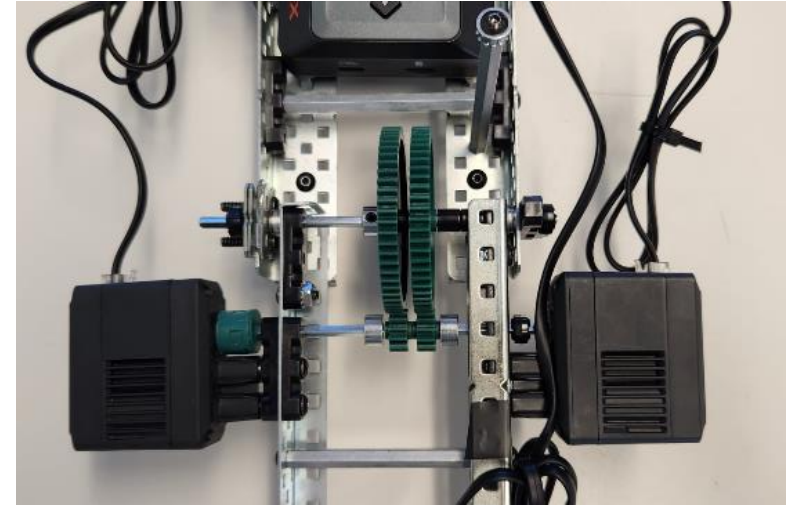


→ Successful run-through with no gap and manually pressing switches

Design Process

Issues + Solutions

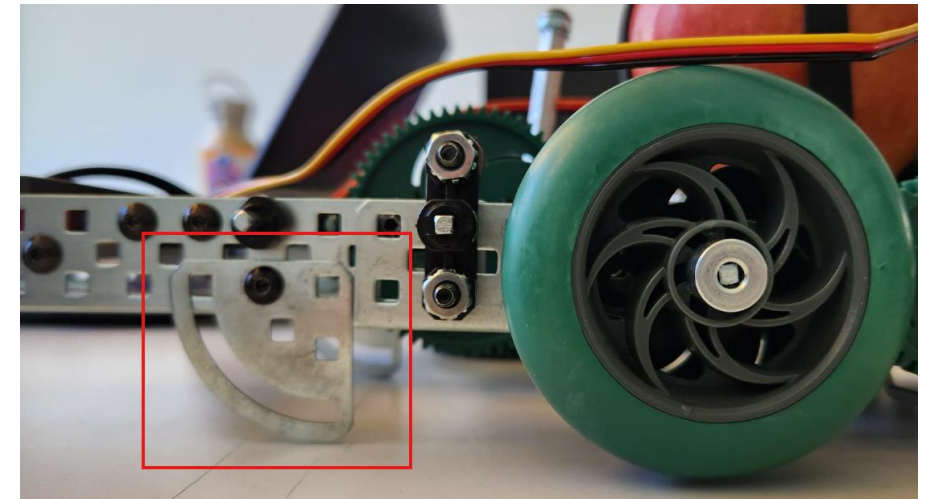
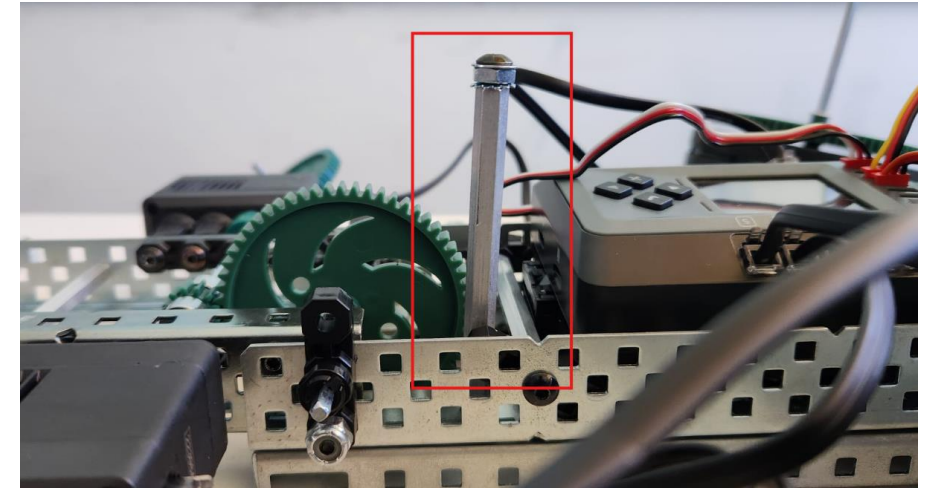
Issue	Root Cause	Solution
Rear arm not retracting	Interference between arm and chassis	Change C-channel to L-channel
	Insufficient motor torque	Drive shaft with 2 motors
System stopping before gap cleared	Limit switch bouncing during deployment triggers ISR	Change from limit switch to ultrasonic sensor
Gears free-spinning	Drive shaft lock failure	Made our own parts



Design Process

Issues + Solutions

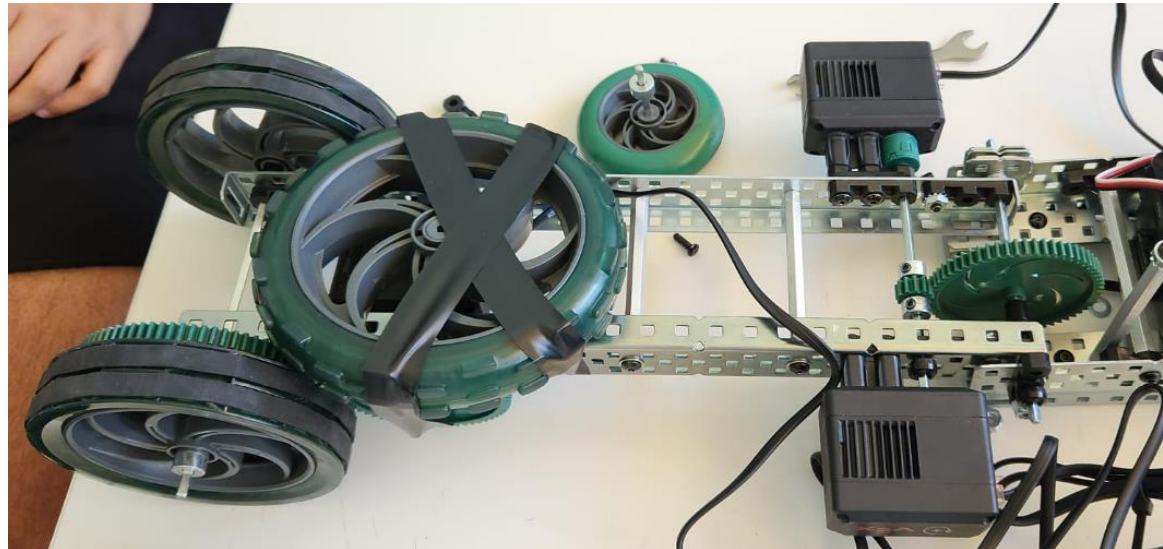
Issue	Root Cause	Solution
Arms retracting too much	Backlash in motor leads to imprecise positioning, so we overshoot	Add mechanical stop (stand-offs)
Main body not clearing the gap	Passive wheels getting caught	Remove free-spinning wheels, add skids and bumpers



Design Process

Optimization

- First iteration was designed to be a platform for future optimization
 - Extra arm torque
 - More rigid than strictly necessary
- Once we proved we could cross a gap, we began extending arms for larger gap-crossings



Retrospective

What worked well?

- Platform design – future-proofing
- Ideate, Test, Select workflow
- Build in parallel – with feedback
- Trade-off: Performance vs. Risk
 - Knowing when to freeze design

What could be improved?

- Lighter design
- Material optimization
- Better use of gear ratios
- System reiteration
 - Redesign to reduce bandaid solutions

